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GST 104: History and Philosophy of Science - How Knowledge Actually Grows

From observation and falsification to African science and the ethics of discovery

GST / COMPULSORY

GST & Compulsory Courses

General Studies (GST / CED)
GST Directorate | 100L

Written for students, by students.

Practical notes, worked examples, practice questions and a revision plan.

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Why this pack exists

This pack explains what makes science different from mere belief, how scientific ideas have changed across history, and how African knowledge systems contributed to medicine, metallurgy and agriculture. It also gives you the vocabulary for ethics-of-science questions.

This is a student-written study companion, not a reproduced lecture handout or a leaked examination paper. Use it to organise the topics you are taught, practise the kind of thinking your course rewards, and walk into class with better questions.

What you will be able to do

- Explain the scientific method, hypothesis, falsifiability, paradigm and theory without collapsing them into one word.
- Trace major shifts from ancient to modern science with accurate examples.
- Describe African contributions to science and technology using specific evidence.
- Discuss ethical issues such as research misconduct, informed consent and environmental responsibility.

How to use this pack

1. Skim the whole pack once in 20 minutes so you know what is inside.
2. Work through one section at a time with your lecture note and recommended text beside you.
3. Attempt the practice questions before reading the answers. Mark what you missed.
4. Use the final checklist to plan your revision week, one tick per study session.

> **Free to download, always**
> This pack is free. There is no fee, no subscription and no payment step before download, and the full PDF is not watermarked or locked. Share the link with a coursemate instead of forwarding the file.

> **Confirm your official course outline**
> Course codes, credit units and outlines can differ by department and session. Treat this pack as supplementary support and confirm the current outline with your lecturer, course rep or departmental handbook before an assessment.

What science is - and is not

Science is a method of producing reliable knowledge through observation, measurement, hypothesis testing and public criticism. It is not a fixed body of facts, and it is not the same as technology.

A hypothesis is a testable prediction. A theory is a well-supported explanatory framework that has survived repeated testing. A law describes a regular relationship. Colloquial English uses 'theory' to mean a guess; science does not.

Falsifiability, associated with Karl Popper, means a claim must specify what evidence would count against it. 'Everything happens for a reason' is not scientific because no observation could refute it.

- Observation and measurement: record what happens under controlled conditions.
- Hypothesis: a specific, testable prediction.
- Experiment: manipulate variables while controlling others.
- Peer review: allow others to check methods, data and reasoning.
- Replication: repeat the study to see whether the result holds.

> **Correlation is not causation**
> Two things rising together may share a third cause, or one may cause the other in the opposite direction. Before claiming causation, look for controlled evidence and a plausible mechanism.

A short history of scientific change

Science has not progressed by simply accumulating facts. It has changed when old frameworks could no longer explain new observations.

- Ancient Greek natural philosophy: Aristotle's physics and the geocentric model.
- Islamic Golden Age: algebra, optics, hospitals and the preservation of Greek texts.
- Copernican revolution: the sun-centred model replaced geocentrism.
- Galileo and Newton: mathematised motion, telescopes and laws of gravitation.
- Darwin and Wallace: evolution by natural selection.
- Einstein: relativity transformed space, time and gravity.
- Watson, Crick, Franklin and Wilkins: the structure of DNA and modern molecular biology.

African science and technology

African societies produced empirical knowledge and technical innovation long before European contact. Recognising this is not sentiment; it is historical accuracy.

- Metallurgy: Nok iron smelting and Igbo-Ukwu bronze casting.
- Medicine: herbal pharmacopoeias, bone setting and knowledge of plant properties, some of which inform modern drug discovery.
- Agriculture: mixed cropping, soil management, terracing and indigenous seed selection.
- Mathematics: fractal-like patterns in African textiles and architecture; counting and weight systems.
- Astronomy: navigation and calendar systems based on celestial observation.

> **Avoid both extremes**
> Do not claim that all modern science was stolen from Africa, and do not claim that Africa had no science. Make specific, evidence-based claims about specific technologies.

Science, technology and society

Scientific knowledge is never automatically good. Its application depends on incentives, regulation and who bears the risk.

- Research ethics: informed consent, confidentiality, honest reporting and approval by an ethics committee.
- Misconduct: fabrication, falsification and plagiarism destroy trust and can harm patients.
- Dual use: the same research can produce medicine or weapons.
- Environmental justice: communities may bear environmental costs without sharing the benefits.
- Access: a drug that exists but is unaffordable is not a solution for the person who needs it.

What makes an explanation scientific

Science is not a body of facts but a disciplined way of testing explanations. The core requirement is falsifiability: a scientific claim must rule something out, so that evidence could count against it. "All metals expand when heated" rules out a metal that contracts; "everything happens for a reason" rules out nothing and is therefore not a scientific claim, however true it might feel.

The hypothetico-deductive pattern runs: observe a puzzle, propose a hypothesis, derive a prediction that would be surprising if the hypothesis were false, test it, and revise. The prediction step is where weak work fails, because a prediction that the hypothesis does not actually require cannot test it.

Confirmation is cumulative but never final. A theory survives because it has repeatedly made risky predictions that came true, not because it has been proved. This is why scientists speak of the best-supported explanation rather than the truth, and why a single decisive anomaly can be more valuable than a hundred confirmations.

Kuhn, Popper and the question of change

Popper argued that science progresses by conjecture and refutation: we propose bold theories and try to destroy them, and what survives is provisionally accepted. Kuhn argued that this describes crises rather than normal work. In his account, a scientific community operates within a paradigm - shared assumptions, methods and exemplars - and normal science is puzzle-solving within that paradigm. Anomalies accumulate, and when the paradigm can no longer absorb them, a revolution replaces it.

The two accounts answer different questions. Popper explains the logic of justification; Kuhn explains the sociology of change. A strong exam answer uses both without collapsing them: Popper tells you what should persuade a scientist, Kuhn tells you what actually does.

The paradigm debate raises a real problem: if communities with different paradigms cannot agree on standards, is science rational? The usual reply is that paradigm choice is constrained by predictive success, experimental reproducibility and practical application - not by arbitrary preference.

> **Criticism is the method**
> A claim that cannot be criticised cannot be tested. When you meet an unfalsifiable claim, the question is not whether it is comforting but what evidence would change the holder's mind.

Science, technology and society in Africa

The history of science is not a European monopoly narrative. Metallurgy, medicine, astronomy, agriculture and mathematics developed in many centres, and the history of their transfer matters: colonial science in Africa was largely organised around extraction, survey and tropical medicine for the benefit of administrators, which shaped institutions long after independence.

Indigenous knowledge systems contain empirical content - soil classification, plant medicine, weather prediction - that is testable and has sometimes been validated by laboratory work. Treating such knowledge as either sacred or worthless are both errors; the useful question is what claims it makes and how they can be checked.

Contemporary debates about research funding, brain drain and technology transfer are part of this history rather than a break from it. A good answer links the philosophy of science to the conditions under which science is actually done.

WORKED EXAMPLES

Do it once, slowly

These are not model answers to copy. They show the reasoning a strong student would use, step by step, so you can repeat the process on a question you have not seen.

1. Testing a hypothesis properly

A student claims that studying in groups improves examination performance. Build a testable version.

1. State the hypothesis with a direction: students who study in groups of three to five score higher than students who study alone.
2. Define the measure: mean score on the same examination.
3. Define the comparison group and how students are assigned.
4. Identify confounds: prior ability, attendance, time spent, subject difficulty.
5. State what result would count against the hypothesis.
6. Acknowledge limits: one cohort, one institution, no claim beyond it.

> **What to carry forward**
> A hypothesis that cannot fail is not a hypothesis. Name the disconfirming result in advance.

2. Answering "is astrology a science?"

A classic exam question. Avoid the trap of asserting an answer without criteria.

1. State criteria first: testability, predictive precision, falsifiability, openness to revision.
2. Apply them: astrology makes predictions but typically vague, post-hoc adjusted, and unfalsifiable in practice.
3. Acknowledge the historical point: astronomy and astrology were once intertwined.
4. Distinguish the claim from its practitioners - sincerity is not evidence.
5. Conclude that it fails the falsifiability criterion, not that it is meaningless to its users.

> **What to carry forward**
> Answer with criteria, apply them, then conclude. The criteria earn the marks.

EXAM TECHNIQUE

Questions with model answers

Written in the shape your course is usually examined, with the marking logic spelled out. Attempt each one under time before you read the outline.

Q1. Explain falsifiability and why it is central to the demarcation of science.

15 marks

1. Define: a claim must forbid some observable state of affairs.
2. Explain the demarcation problem it solves.
3. Give one falsifiable and one unfalsifiable example.
4. Note the objection: some good science is not directly testable, requiring auxiliary assumptions.
5. Conclude on fallibilism rather than certainty.

Q2. Compare Popper and Kuhn on scientific change.

20 marks

1. Popper: conjecture and refutation, permanent openness to criticism.
2. Kuhn: paradigms, normal science, anomaly, revolution.
3. Show how each accounts for a historical episode.
4. Explain what each explains that the other struggles with.
5. Conclude that they address justification and sociology respectively.

> **Where marks are lost**
> Presenting Kuhn as "irrational" or Popper as "naive". Both positions are more careful than that.

REFERENCE

Glossary

Use these definitions in your own sentences; a copied definition rarely earns full marks.

Falsifiability. The property of a claim that allows evidence to count against it.

Paradigm. The shared assumptions, methods and exemplars of a scientific community.

Normal science. Puzzle-solving within an accepted paradigm.

Anomaly. An observation a paradigm cannot explain, prompting revision.

Hypothesis. A testable proposed explanation derived from a theory.

Confirmation. Evidence consistent with a hypothesis, always provisional.

Demarcation. The problem of distinguishing science from non-science.

Induction. Generalising from observed cases to a wider claim.

Deduction. Deriving specific predictions from general principles.

Reproducibility. The ability of others to obtain the same result by the same method.

Auxiliary assumption. A background belief required for a test, which may itself fail.

Indigenous knowledge. Locally developed empirical knowledge about environment, health and production.

Your four-week plan

Week 1 - Method

Falsifiability.

- Write three falsifiable and three unfalsifiable claims.
- Design a test for one falsifiable claim.
- Name the disconfirming result.

Week 2 - Change

Popper and Kuhn.

- Summarise each in 150 words.
- Apply both to one historical episode.
- List one strength and one weakness of each.

Week 3 - Context

Science and society.

- Trace one colonial scientific institution.
- Evaluate one indigenous knowledge claim.
- Link funding to research priorities.

Week 4 - Essay

Timed answer.

- Outline a 20-mark answer.
- Write it in 35 minutes.
- Check that criteria are stated before applied.

Where to read more

Entry-level texts you can find in the university library or a departmental reading list. Confirm editions with your lecturer.

- Karl Popper, Conjectures and Refutations - the opening chapters on falsification.
- Thomas Kuhn, The Structure of Scientific Revolutions - chapters on normal science and revolutions.
- An introduction to the history of science in Africa covering indigenous knowledge and colonial institutions.

PRACTICE, NOT PREDICTION

Check your understanding

These questions are written for this pack to test the ideas above. They are not past examination questions and do not predict what your lecturer will set. Answer first, then compare.

1. What makes a claim falsifiable?

Answer: It specifies observations that would count against it. If no possible evidence could refute the claim, it is not scientifically testable.

2. What is the difference between a hypothesis and a theory?

Answer: A hypothesis is a specific testable prediction. A theory is a broad explanatory framework supported by many lines of evidence.

3. Give one African contribution to metallurgy.

Answer: The Nok culture produced iron smelting evidence dating to roughly the first millennium BCE; Igbo-Ukwu is known for sophisticated bronze casting.

4. Why is peer review important?

Answer: It exposes methods and claims to independent scrutiny before publication, reducing the chance that errors or weak reasoning go unchallenged.

5. Explain the difference between correlation and causation with an example.

Answer: Ice-cream sales and drowning deaths both rise in hot weather, but ice cream does not cause drowning. Heat is a common cause.

6. Name two ethical requirements for research involving human participants.

Answer: Informed consent and ethics committee approval; confidentiality and the right to withdraw are also essential.

Your 7-step action plan

Tick each step only when you have done it. A tick you cannot defend is worse than an empty box.

- [ ] Write definitions of hypothesis, theory, law and paradigm in your own words.
- [ ] Create a timeline of six scientific shifts with one key figure each.
- [ ] List four African technologies and the evidence that supports them.
- [ ] Find one news report that confuses correlation with causation and rewrite the headline accurately.
- [ ] Write a short paragraph on why falsifiability matters.
- [ ] Explain informed consent to a friend in under one minute.
- [ ] Review one unethical research case and identify what should have been done differently.

Keep going

One pack will not carry a semester. Pair this with your lecture notes, tutorials, and the past questions your department actually releases. If a topic still feels unclear, take a specific question to your lecturer or course rep - "I do not understand the difference between X and Y" gets a better answer than "I do not understand this course".

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YOUR WORKSPACE

Notes, questions and things to ask

Use this page in class or during revision. Write questions you could not answer instead of copying slides.

Three ideas I can now explain without notes:

A question I still cannot answer:

For my lecturer or course rep:

Next revision action and date:

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